

EXTRACTION PIPELINE

— BUILT WITH VISION



DOCLING ML

● WEDNESDAY PULSE · WEEK 30



Wednesday Pulse

PIPELINE

need 99% ACCURACY

Manas

ytlc

15 July → 22 July 2026

WHAT I BUILT

A complete PDF-to-JSON extraction pipeline for NCERT Class 11 & 12 textbooks, designed to produce structured training data for LLMs.

HOW IT WORKS

- 1. PDF Ingestion:** PDFs are uploaded via FastAPI REST endpoints or processed in batch from a directory.
- 2. Page Rendering:** Each page is converted to high-res PNG using PyMuPDF (fitz) for downstream processing.
- 3. Text Extraction:** Two strategies available — PyMuPDF for clean text-based PDFs, and Tesseract OCR for scanned pages.
- 4. Content Structuring:** Physics-specific structurer parses raw text into sections, examples, exercises, in-text questions, tables, and LaTeX equations using regex patterns.
- 5. Image Processing:** OpenCV removes NCERT watermarks from figures; figures are extracted and linked to captions.
- 6. Output:** Clean JSON with book metadata, structured content, and statistics — ready for PostgreSQL storage or direct LLM training.

TECH STACK

Backend: Python, FastAPI, Pydantic | **PDF:** PyMuPDF, Tesseract, Poppler | **AI:** Docling ML | **Images:** OpenCV | **DB:** PostgreSQL with JSONB

● GREEN FLAG — PIPELINE BUILT

Multi-strategy extraction with PyMuPDF and Tesseract OCR.

- Physics structurer: sections, tables, examples, equations
- 7 physics + 14 biology chapters extracted
- FastAPI web interface with REST endpoints

● IN PROGRESS — DOCLING ML

Integrating Docling's ML layout model for better section/table/figure detection.

- Figure extraction with caption-anchored cropping
- Unified structurer for cross-subject patterns

● RED FLAG — ACCURACY ~70-80%

Current extraction not at 99% target. Errors propagate to LLM training.

- Tables lose alignment, equations truncated
- Bio/Chem/Maths need dedicated structurers
- **Ignoring i3w IL backend for now** — focusing entirely on extraction accuracy